

Assignment 1: Forecasting Yields and Inflation

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Introduction

We propose an alternative model to the C.S. approach to estimate bonds yields, building on Favero, Melone and Tamoni (2022). Specifically, following our reference, we try to address the forecasting of bonds yields by directly modeling the drift in bonds yields. This approach turns out to be successful because, as opposed to V.A.R. or C-V.A.R., employs information on slow moving parameters of the economy such as the *Middle to Young* ratio and the growth rate of the flexible price equilibrium. We consider these long-term data as exogenous processes, thereby enhancing our ability to capture the underlying stochastic trends in bond yields over the long-term.

Is Co-integration Enough?

Cointegration is fundamentally argued to be crucial for avoiding the differentiation of variables, thereby preserving information about their levels. It is important to note, however, that cointegration typically provides information only about the relative levels of two variables, and the primary work is still accomplished by a process that determines the drift of the co-moving variables.

Starting from the C.S. approach¹, we observe that

$$Y_t = \theta(1 - \delta) \sum_{i=0}^{\infty} \delta^i E_t y_{t+i} + c \quad (1)$$

Eq. (1) implies that:

$$S_t = E_t S_t^* + c, \quad S_t^* = \theta \sum_{i=1}^{\infty} \delta^i \Delta y_{t+i} \quad (2)$$

Define $S_t := Y_t - \theta y_t$ and subtract both sides of eq (1) by θy_t , we obtain that:

$$\begin{aligned} Y_t - \theta y_t &= \theta(1 - \delta) \sum_{i=0}^{\infty} \delta^i \mathbb{E}_t [y_{t+i}] + c - \theta y_t \\ S_t &= -\theta \delta y_t + \delta \theta y_{t+1} + \theta \sum_{i=2}^{\infty} \delta^i \mathbb{E}_t [\Delta y_{t+i}] + c \\ S_t &= \theta \sum_{i=1}^{\infty} \delta^i \mathbb{E}_t [\Delta y_{t+i}] + c \end{aligned}$$

¹We take their approach in the more general form as it's easier in terms of computations and it's still flexible enough to be applied both to stocks and to the term structure.

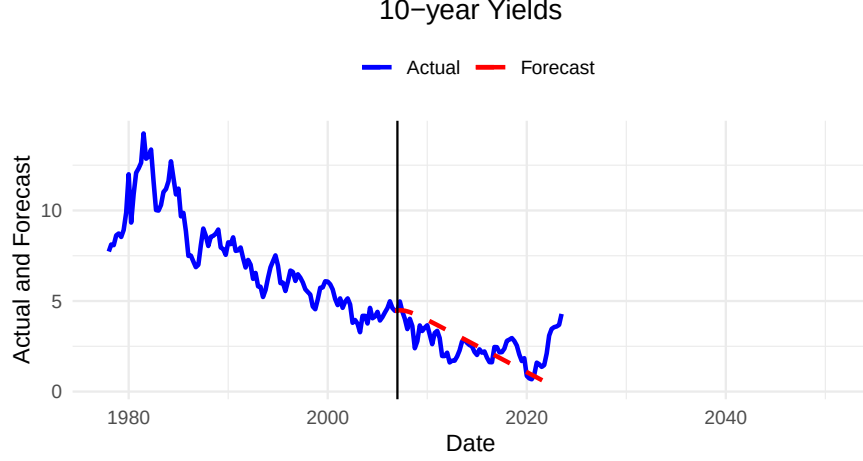


Figure 1: 10 years out of sample forecasts vs actual yields

Previous considerations allow us also to link the spread to ΔY_{t+1} :

$$S_t = \mathbb{E}_t \left[\theta \delta \sum_{i=0}^{\infty} \delta^i y_{t+i+1} \right] - \mathbb{E}_t \left[\theta \delta \sum_{i=0}^{\infty} \delta^i y_{t+i} \right] + c$$

And hence we find that:

$$S_t = \mathbb{E}_t \left[\frac{\delta}{1-\delta} \Delta Y_{t+1} \right] + c \quad (3)$$

If y_t is integrated of order 1, then Δy_t is stationary. Eq. (2) implies that also S_t is stationary. By means of eq. (3) then also ΔY_{t+1} is stationary. Thus one can use S_t and Δy_t , or S_t and ΔY_t , as stationary variables that summarize the bivariate history of y_t and Y_t . (The pair Δy_t and ΔY_t is also stationary, but by using these one would lose information on the relative levels of y_t and Y_t) Previous considerations suggest a model of the form:

$$\begin{bmatrix} \Delta y_t \\ S_t \end{bmatrix} = \begin{bmatrix} a(L) & b(L) \\ c(L) & d(L) \end{bmatrix} \begin{bmatrix} \Delta y_{t-1} \\ S_{t-1} \end{bmatrix} + \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix}$$

We hence estimate a $p = 1$ V.A.R. of the form:

$$\begin{bmatrix} \Delta y_t \\ S_t \end{bmatrix} = \begin{bmatrix} -0.17234 & 0.22254 \\ 0.10250 & 0.73446 \end{bmatrix} \begin{bmatrix} \Delta y_{t-1} \\ S_{t-1} \end{bmatrix} + \begin{bmatrix} -0.47105 \\ 0.46816 \end{bmatrix} \quad (4)$$

Once we have estimated the VAR we can proceed to construct our forecasts of Δy_t and of S_t . We can then forecast T-bonds at 10-year maturity by simply forecasting the 3m yield recursively as follows:

$$i_{t+1}^{3m} = i_t^{3m} + \Delta y_{t+1} \quad (5)$$

And then recursively:

$$i_{t+1}^{10y} = i_t^{10y} + S_{t+1} \quad (6)$$

Some Considerations

Forecasting with a C-Var comes with the drawback of not having explicitly modeled the unit root process, which, in this context, is represented by S_t . For one-step-ahead forecasts, C-Var is effective as it incorporates the current level of S_t . However, when extending the forecast to multiple steps ahead, it becomes necessary to also predict future values of S_t .

The issue arises because in this specification we are essentially forecasting the stochastic trend with a deterministic (linear) trend, then cointegration constrains the value of Δy_t to this trend.

This limitation is explicit in the figure showing that forecasts tend to "linearize", potentially dropping below zero with longer forecast horizons.

A simple way to see this process is to take the forecast for a unit root process such as the random walk. The s step ahead forecast is given by:

$$\hat{y}_{t+s} = s\delta + y_t$$

which is a linear trend. If we want to improve the performance of the model in terms of root mean square forecast error (RMSFE) we should be explicit about the modeling of the trend in yields. This can be done by estimating the trending component of yields as based on an appropriately defined interest rate rule.

There is No Harm in Trying

In our forecasting process, we initiated our estimation with a Vector Autoregression (VAR) model, a statistical tool that captures the linear interdependencies among multiple time series by modeling each variable as a linear function of its own past values and those of other variables in the system. To begin, we determined the optimal order (p) of the VAR model and estimated the coefficients using our estimation sample. This estimated VAR model was then employed to generate forecasts for our forecast sample. A general VAR(p) model:

$$y_t = \phi_0 + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + \epsilon_t$$

To identify the most effective lag order for our VAR model, we utilized information criteria, specifically the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). These criteria helped us balance model complexity with fit, ensuring that we selected an optimal lag length that neither underfits nor overfits our data.

Following the model specification and estimation, we evaluated the out-of-sample (OOS) performance of our VAR model. This step was crucial for understanding how well our model could predict future values based on data it had not been trained on.

Lastly, to quantify the accuracy of our forecasts, we calculated the Root Mean Squared Forecast Error (RMSFE). This metric provided a clear measure of the average magnitude of our forecast errors, offering valuable insights into the overall effectiveness of our forecasting process using the VAR model.

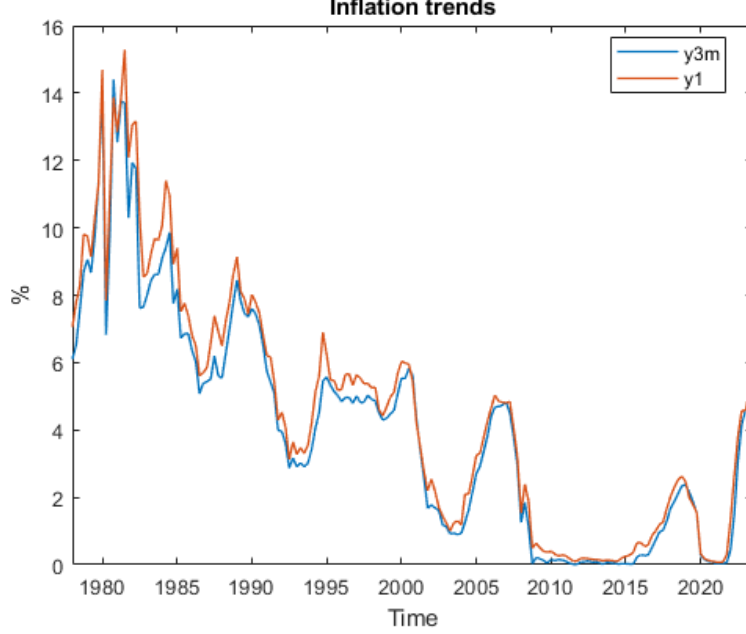


Figure 2: Yield and inflation trend

Modeling The Drift In Monetary Policy

As we have largely discussed in the previous sections, we need a model for the drift in monetary policy. Drawing from basic macroeconomic principles, we can define the natural level of interest rate as follows:

$$r_t^n := \gamma + \sigma \mathbb{E}_t [\Delta y_{t+1}^{\text{flex}}] = \gamma + \sigma \mathbb{E}_t [\Delta y_{t+1}^{\text{natural}}] \quad (7)$$

In its baseline specification, largely influenced by *R.B.C.* economists, the model posits that technology is the sole stochastic factor, leading to the following equality:

$$r_t^n \equiv \gamma + \sigma \mathbb{E}_t \Delta y_{t+1}^n = \gamma + \frac{\sigma(1+\varphi)}{\sigma + \varphi} \mathbb{E}_t \Delta a_{t+1}$$

Once we allow for capital to have a role and for preference shock hitting the *consumption/savings* decision, the output growth will depend on the level of capital and on technology.

As known, the optimal policy rule is achieved when in:

$$i_t = \gamma + \phi_\pi \pi_t + v_t$$

$\phi_\pi \rightarrow \infty$, which is equivalent to the following interest rate rules (that, as the previous one, ensure determinacy and uniqueness of equilibrium under specific conditions on the coefficients²):

$$i_t = r_t^n + \phi_\pi \pi_t + \phi_x x_t + v_t \quad (8)$$

²See Blanchard Khann conditions.

$$i_t = r_t^n + \phi_\pi \mathbb{E}_t [\pi_{t+1}] + \phi_x \mathbb{E}_t [x_{t+1}] + v_t \quad (9)$$

Allowing for the *steady state* inflation to be different from zero and recognizing that the C.B. suffers from a trade off between inflation stabilization and efficiency in the presence of cost push shocks, we use a general (N.K. since forward-looking) interest rate rule that we borrow from Favero, Melone and Tamoni (2022):

$$i_t = i_t^* + \beta_1 \mathbb{E}_t (\pi_{t+1} - \pi_{t+1}^*) + \beta_2 \mathbb{E}_t (x_{t+1}) + u_t \quad (10)$$

Where i_t is the policy instrument that we proxy with $T - Bills$ at 3-months maturity. The interest rate so follows a standard Taylor rule, incorporating a key principle from the classical New Keynesian (N.K.) paradigm: it is formulated in terms of expectations about the future. Before we start to explain the strategy to estimate the natural interest rate, the real driver of the model, we need to explain how we intend to estimate eq. (10). As in our reference we use lags of inflation and output gap as instrumental variables and as well the following V.A.R. allow us to project the variables O.O.S.:

$$\begin{aligned} (\pi_t - \pi_t^*) &= \theta_{1,1} (\pi_{t-1} - \pi_{t-1}^*) + \theta_{1,2} x_{t-1} + \theta_{1,3} (i_{t-1} - i_{t-1}^*) + v_{1,t} \\ x_t &= \theta_{2,1} (\pi_{t-1} - \pi_{t-1}^*) + \theta_{2,2} x_{t-1} + \theta_{2,3} (i_{t-1} - i_{t-1}^*) + v_{2,t} \end{aligned}$$

The Driver of Monetary Policy

As hinted in the previous sections, the main idea is to model the interest rate rule by exploiting the fact that we have informations (in the form of long-term forecasts) about its determinants. Favero, Melone and Tamoni (2022) use:

$$i_t^* = \gamma_1 MY_t + \gamma_2 \Delta x_t^{pot} + \gamma_3 \pi_t^* \quad (11)$$

As we have seen in eq (7) the natural interest rate is a function of the natural growth rate of the economy. If we allow for capital in the economy, also shocks to capital accumulation influence the flex-price equilibrium. Favero, Melone and Tamoni (2022) proxy shocks to the accumulation of capital with the *middle to young* ratio. This represents a sensible view, that we share, that capital accumulation is influenced by the age structure of the population. We would expect indeed that MY_t is negatively related to the equilibrium rate in the economy. Indeed if $MY_t \uparrow$ supply of capital increases and hence the equilibrium rate in the economy becomes lower. This is a results that is confirmed once we estimate (10) by instrumental variables.

The Yield Curve

Given our primary interest in long-term forecasting of the *yield curve*, we are now interested in how to transform the modeling of the policy instrument into a model for yields. Borrowing from the *E.T.*, we can

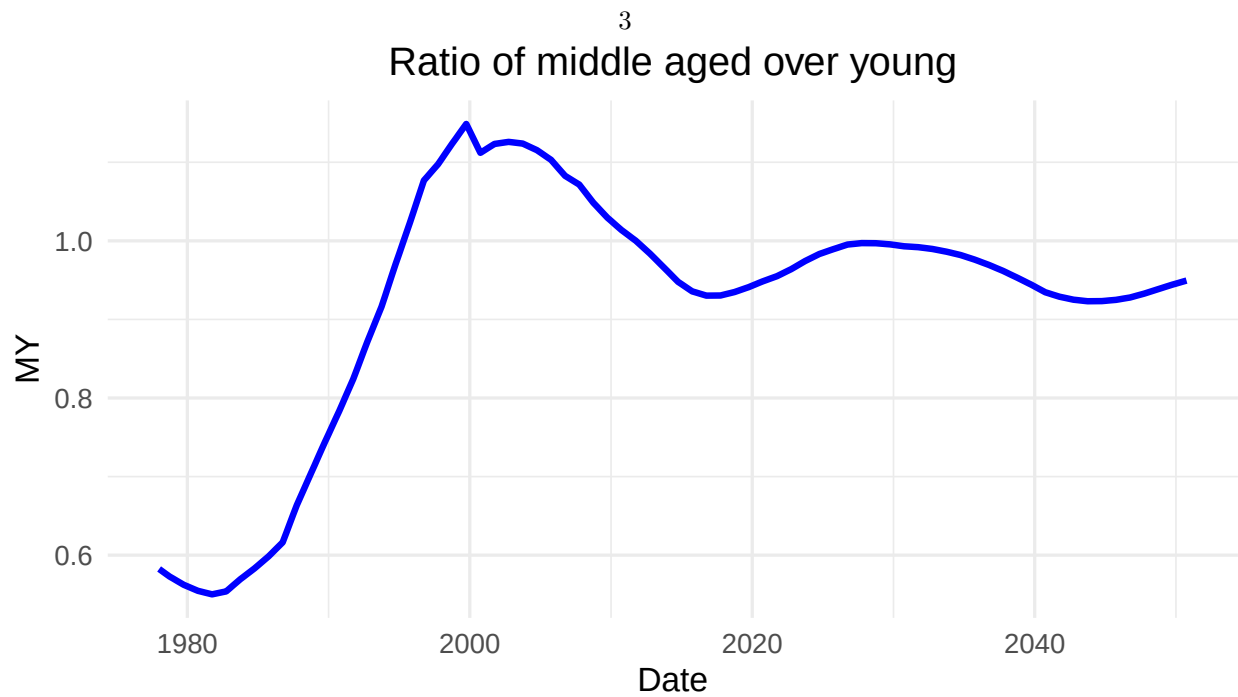


Figure 3: Ratio of middle aged to young

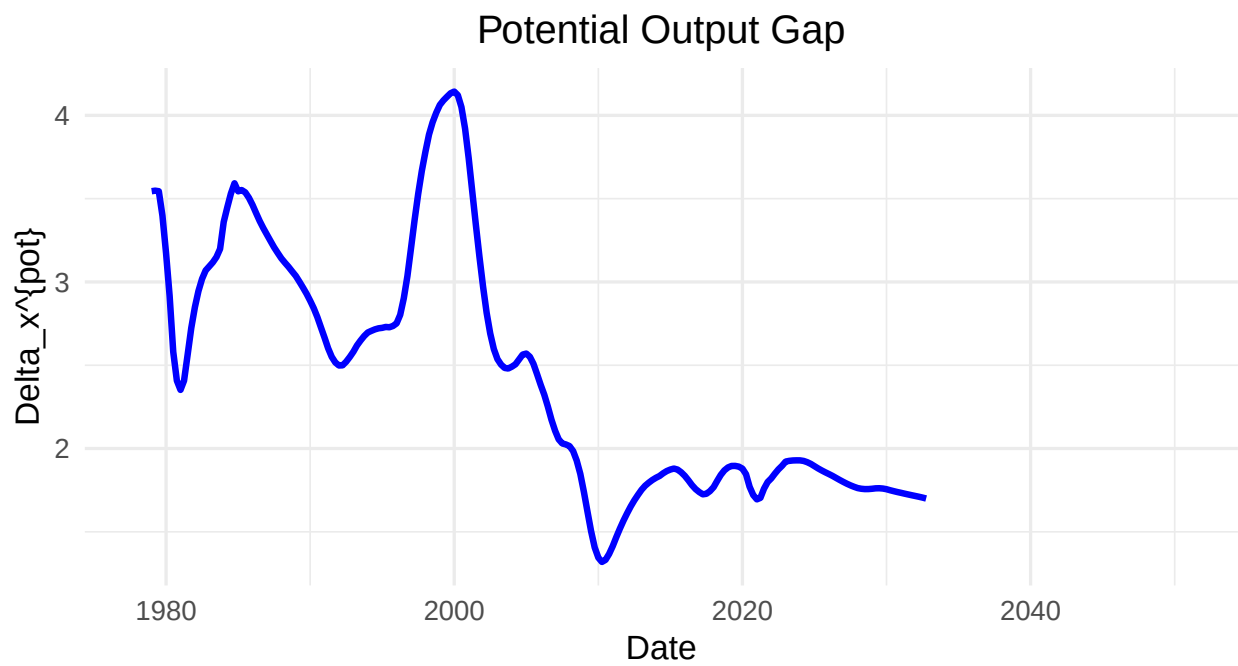
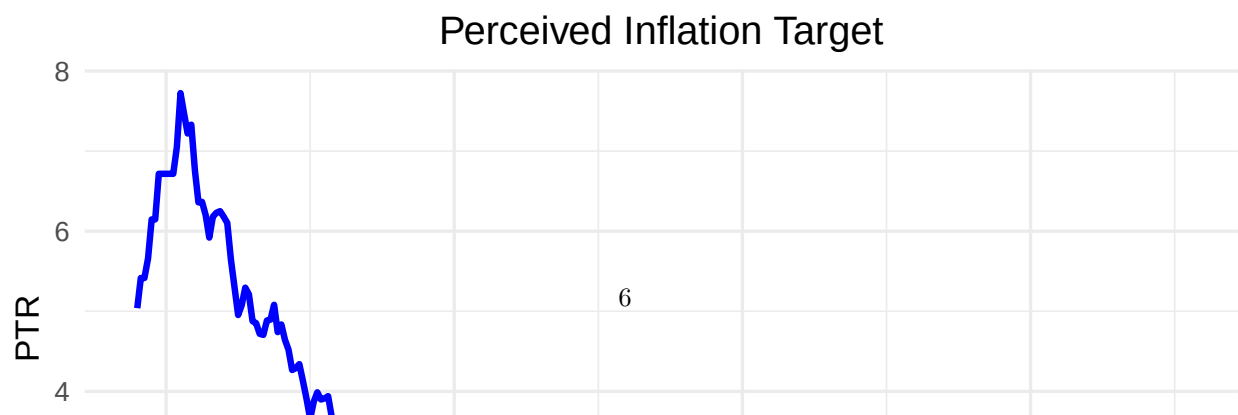


Figure 4: Potential Output Gap



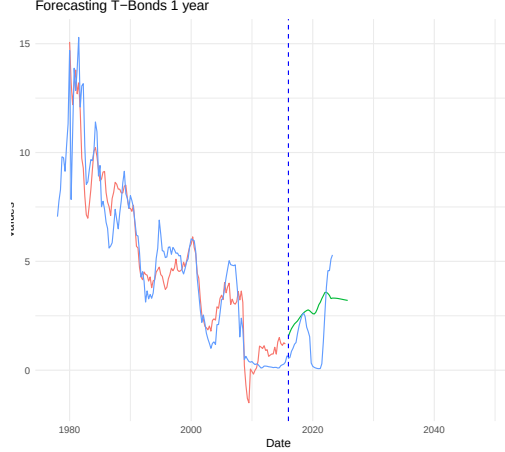


Figure 6: In Sample Performance. Blue line: actual 3 months yields, green line: O.O.S. predictions, red line: in sample forecast

obtain, as suggested by the original authors Favero, Melone and Tamoni (2022):

$$\begin{aligned}
 y_t^n &= y_t^{(n),*} + \delta_0^n + u_t^n \\
 y_t^{(n),*} &= \left(\frac{1}{n}\right) \sum_{i=0}^{n-1} E_t \left[y_{t+i}^{(1)} \right] \\
 u_t^{(n)} &= \rho^{(n)} u_{t-1}^{(n)} + \epsilon_t^{(n)}
 \end{aligned}$$

The scope of this equation is much larger than the use we employed in this exercise. In the original paper, the authors are indeed interested also in analyzing the *cycle* and to which extent such deviations have a R.E. interpretations or are the results of behavioural biases³. We will elude from the modeling of the cycle and we focus solely on the trends. In our context, it is fundamental the fact that, as one would expect, log returns over n-period horizon should be (at the exception of a constant or a stationary risk premium) driven by the expectations of the the sum of these returns over the holding period. However, it is important to note that they are not imposing a N.A. condition; thus, deviations from the trend are still possible.

Estimation

We estimate eq. (10) by means of a regression with instrumental variables and we obtain that:

$$i_t = -2.28^{***} MY_t + 0.94^{***} \Delta_x^{pot} + 1.52^{***} \pi_t^* + 0.52^{***} E_t (\pi_{t+1} - \pi_{t+1}^*) + 0.35^* E_t (x_{t+1}), \quad R^2 = 96\%$$

As we were expecting there is a negative relationship, *ceteris paribus* with the MY_t ratio. As an anomaly, while all the coefficients are comparable till decimals points, we have a non negligible change in the coefficient multiplying the inflation gap: in the original paper that value is 0.709. Using the V.A.R. that we estimated before we are now able to project O.O.S. the policy rate obtaining the trend in Figure 6.

Having obtained the policy rule, and always using the V.A.R. for projections we can extract trends for the t-bonds at 1 year and 10 year maturity.

³They analyze if diagnostic expectations could predict the large deviations of bond prices over their trend

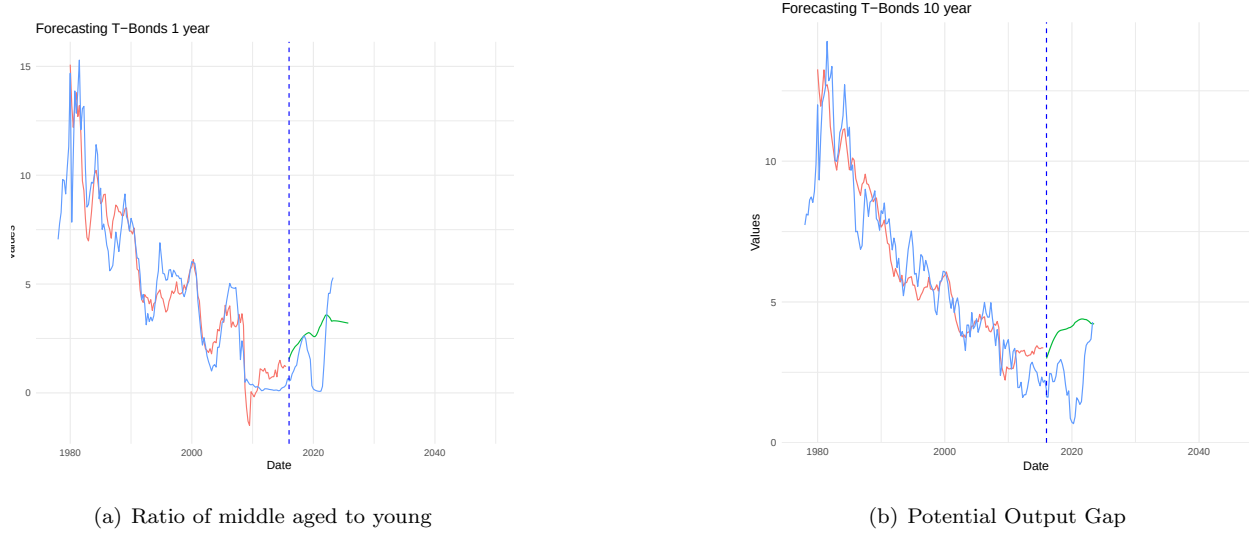


Figure 7: O.O.S. forecasts for T-Bonds at 1 and 10 year maturities

Inflation Forecasting

In the model variables are nested and hence to forecast the interest rate and the T-bonds we still had to compute via:

$$\begin{aligned}
 (\pi_t - \pi_t^*) &= \theta_{1,1} (\pi_{t-1} - \pi_{t-1}^*) + \theta_{1,2} x_{t-1} + \theta_{1,3} (i_{t-1} - i_{t-1}^*) \\
 x_t &= \theta_{2,1} (\pi_{t-1} - \pi_{t-1}^*) + \theta_{2,2} x_{t-1} + \theta_{2,3} (i_{t-1} - i_{t-1}^*)
 \end{aligned}$$

the output gap and inflation gap. Once having this variables it becomes a question of specifying the behaviour of the last *non exogenous* variable, namely π_t^* . We model such variable as a *random walk* for two order of reasons: (1) there is reliable way of estimating this variable and (2) we proxy the inflation target with *PTR long term inflation expectations* and even if there is variability it oscillates around the 2% target.

Some Considerations about Performance

The O.O.S. has been computed from (2016:Q1) to (2025:Q4). Due to the lack of the real world counterpart data we computed the RMSFE over the horizon (2016:Q1) to (2023:Q4). We obtain the values summarized in table

Table 1: <i>Mean Square Forecast Errors</i>		
T-Bonds 1 year	i_t^4	1.19%
T-Bonds 10 years	i_t^{40}	2.03%
Inflation	π_t	2.37%

The model performs fairly well with regard to the policy instrument and with regard to the 1 year T-Bonds. Some doubts could arise on why, even if it's fundamentally based on the policy interest, it predicts a flattening of short rates at the end of the forecasting period when, instead, we know that interest rates

have spiked up. If one believes that we correctly specified the model and we estimated it correctly he should embrace the fact that those are temporal deviations from the trend that will somehow come revert back. Another explanation is that out of sample the model is incapable of realize if permanent shocks to the economy have occurred or not. This is clear if we look at an equivalent formulation of the interest rate model taken from Favero, Melone and Tamoni (2022):

$$y_t^{(1)} = \alpha_1 r_t^* + \alpha_2 \pi_t^* + \beta_1 E_t (\pi_{t+1} - \pi_{t+1}^*) + \beta_2 E_t (x_{t+1}) + u_t^{(1)}$$

$$r_t^* = \gamma_1 MY_t + \gamma_2 \Delta x_t^{pot}$$

The coefficients α_1 and α_2 are used to give flexibility at the central bank when facing shocks. A break in those parameter could well be the reason for inaccurate forecast. What's seems more concerning, regarding the correct estimation of the model, is the T-Bonds 10 years and specifically the concavity that is predicted. This is disturbing to the extent that we accumulate a high *MSFE* even though we could probably forecast the direction of the trend. It's partially caused by the fact that 10 year bonds are a long weighted average with quarterly data and hence they inherit the fact that short term rates (in trend) are predicted to become more flat. Having said so one should carefully check if this is due to errors in the passage from I.S. to O.O.S. which is the critical step of all the model, but specifically in cases of nested models. We conclude with the main point that we have tried to deliver, namely that when facing a stochastic trend one should not disregard the possibility to directly model the drift of such variables using, when possible, slow and deep parameters of the economy to provide long term reliable forecasts.